

OHIO STATE UNIVERSITY, OH, USA

WMO: 724288

Lat: 40.078N

Lon: 83.078W

Elev: 276

StdP: 98.06

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.7	-20.0	0.7	-14.6	-17.5	0.8	-12.0	11.6	1.0	10.6	-1.2	3.3	270	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.4	23.1	31.2	22.7	29.9	22.0	24.8	30.1	24.0	28.9	23.2	27.7	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.4	27.4	22.4	17.7	26.7	21.9	17.2	26.0	76.3	30.1	73.2	29.1	70.2	27.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.7	-18.8	33.9	3.7	1.3	-21.4	34.8	-23.6	35.6	-25.7	36.3	-28.4	37.2		
(5)			-19.2	26.1	3.6	0.9	-21.8	26.7	-23.8	27.2	-25.9	27.7	-28.5	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-1.8	-0.2	4.8	11.5	17.3	21.7	23.5	22.8	19.3	12.5	6.1	0.9	(6)
(7)		DBStd	10.27	6.75	6.28	6.28	5.10	4.62	3.48	2.80	2.75	4.00	4.97	5.20	5.66	(7)
(8)		HDD10.0	1346	369	289	188	42	3	0	0	0	29	138	287		(8)
(9)		HDD18.3	2999	624	518	422	209	77	11	1	2	36	192	367	540	(9)
(10)		CDD10.0	1926	3	4	26	88	230	351	417	396	278	107	22	5	(10)
(11)		CDD18.3	537	0	0	1	6	45	112	160	139	64	11	0	0	(11)
(12)		CDH23.3	4634	0	0	7	55	398	962	1405	1148	572	86	2	0	(12)
(13)	CDH26.7	1446	0	0	1	5	98	313	478	354	179	18	0	0	(13)	
(14)	Wind	WSAvg	3.5	4.3	4.2	4.2	3.4	3.2	2.8	2.7	2.7	3.3	3.7	3.9	(14)	
(15)	Precipitation	PrecAvg	991	68	55	85	92	102	108	98	91	83	69	63	75	(15)
(16)		PrecMax	1246	233	106	162	168	172	217	148	202	142	144	111	118	(16)
(17)		PrecMin	843	29	36	25	50	41	42	39	39	6	31	17	29	(17)
(18)		PrecStd	124	50	21	36	31	40	47	30	45	34	31	25	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	19.4	24.1	27.5	30.8	32.9	33.8	33.3	33.0	29.0	21.9	17.6	(19)
MCWB			13.8	14.9	16.8	18.1	21.3	22.6	24.4	23.4	22.0	20.9	16.1	14.5	(20)	
2%		DB	13.9	15.0	21.1	25.0	29.1	31.5	32.3	31.7	30.9	25.6	18.9	14.4	(21)	
		MCWB	12.1	11.7	14.7	16.2	20.2	22.4	23.9	23.1	21.8	18.6	14.1	12.1	(22)	
5%		DB	11.3	12.3	17.7	22.7	27.6	30.1	31.0	30.1	28.7	23.1	17.1	12.0	(23)	
		MCWB	9.5	9.5	12.6	15.1	19.7	21.7	23.2	22.5	20.9	17.2	13.1	10.1	(24)	
10%		DB	7.8	8.8	15.0	20.6	25.8	28.4	29.4	28.7	26.9	21.1	14.4	9.0	(25)	
		MCWB	5.7	6.0	10.9	14.2	18.8	21.0	22.4	21.7	19.7	16.5	11.1	7.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.6	17.6	19.7	22.8	24.7	26.1	25.5	24.2	21.8	17.3	15.2	(27)
MCDWB			16.0	19.0	23.4	25.7	28.6	30.4	31.7	30.9	30.2	27.3	20.3	17.0	(28)	
2%		WB	12.4	12.5	15.4	17.7	21.7	23.7	25.1	24.5	23.0	20.3	15.4	12.7	(29)	
		MCDWB	14.0	14.2	19.8	22.6	27.5	29.1	30.5	29.5	28.0	23.8	18.0	14.1	(30)	
5%		WB	9.6	9.3	13.7	16.4	20.6	22.9	24.2	23.7	22.0	18.6	13.5	10.3	(31)	
		MCDWB	10.9	12.1	16.7	20.7	25.7	28.1	29.1	28.1	26.5	22.2	16.2	11.8	(32)	
10%		WB	5.9	6.2	11.5	15.1	19.6	22.0	23.4	22.8	21.0	16.8	11.6	7.4	(33)	
		MCDWB	7.9	8.5	14.8	19.3	24.3	26.9	27.7	26.7	24.9	20.4	14.2	9.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.8	8.5	9.9	11.5	11.2	10.6	10.3	10.6	11.5	10.7	9.2	7.3	(35)	
5% DB		MCDBR	10.0	12.6	13.4	13.8	12.9	12.2	11.4	11.7	13.5	13.3	12.5	9.8	(36)	
		MCWBR	8.8	9.9	8.5	8.0	6.1	5.3	4.7	4.9	5.5	7.1	8.7	8.6	(37)	
		MCDBR	9.4	11.7	11.5	11.6	11.0	10.4	10.0	9.6	10.4	11.3	10.6	9.5	(38)	
5% WB		MCWBR	8.9	10.1	8.2	7.7	5.7	5.1	4.6	4.6	4.9	6.5	8.5	9.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.329	0.359	0.395	0.436	0.452	0.460	0.452	0.393	0.362	0.327	0.305	(40)	
taud		2.412	2.337	2.328	2.251	2.176	2.176	2.165	2.168	2.312	2.388	2.447	2.450	(41)		
Ebn at Noon		852	883	892	878	846	830	820	815	844	834	824	830	(42)		
Edh at Noon		82	102	115	133	147	147	147	143	116	96	79	74	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.63	2.50	3.49	4.72	5.49	6.02	5.89	5.42	4.42	3.04	2.04	1.38	(44)	
(45)		RadStd	0.15	0.31	0.32	0.41	0.46	0.42	0.34	0.27	0.39	0.27	0.27	0.15	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+68	(47)

Nomenclature: See separate page